Standardizing Quantitative Visual Stratigraphy for Digital Archiving of Ice Cores

Workflow and Methods Documentation

Authors: Gary Hoehne and Grayton Simanson
Advisors: Curt LaBombard and Richard Nunn
Development Dataset: AH2416 Allan Hills Blue Ice Area Ice Core, Antarctica
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Introduction

Ice cores preserve continuous records of past environmental and climatic conditions through variations in ice structure, chemistry, and physical properties. Visual stratigraphy represents one of the most fundamental datasets collected from ice cores, documenting visible characteristics including annual layering, bubble structure, fractures, inclusions, dust horizons, melt features, and other englacial structures.

These observations provide important context for interpreting complementary ice core measurements, including electrical conductivity measurements (ECM), stable isotope analyses, geochemical measurements, and hyperspectral imaging (HSI).

Despite its importance, standardized high-resolution visual documentation is not consistently available for all archived ice cores. While analytical approaches such as ECM and HSI provide valuable physical and spectral characterization, these methods require specialized instrumentation that may not be available at all ice core repositories.

This workflow presents a low-cost, reproducible approach for digitally documenting ice cores using commercially available digital photography equipment and open-source image processing software. The workflow was developed using archived AH2416 ice core sections from the Allan Hills Blue Ice Area, Antarctica, and integrates:

- Standardized visual stratigraphy logging
- Dual-illumination digital imaging
- RAW image processing
- Image stitching and mosaic generation
- Quality control procedures
- Metadata management and archival preparation

Although developed using archived ice core material, this workflow is designed to be transferable to both archived and newly collected ice cores.

The objective of this workflow is to create spatially continuous, depth-referenced digital visual records that preserve ice core observations for long-term curation, future analysis, and integration with complementary datasets.

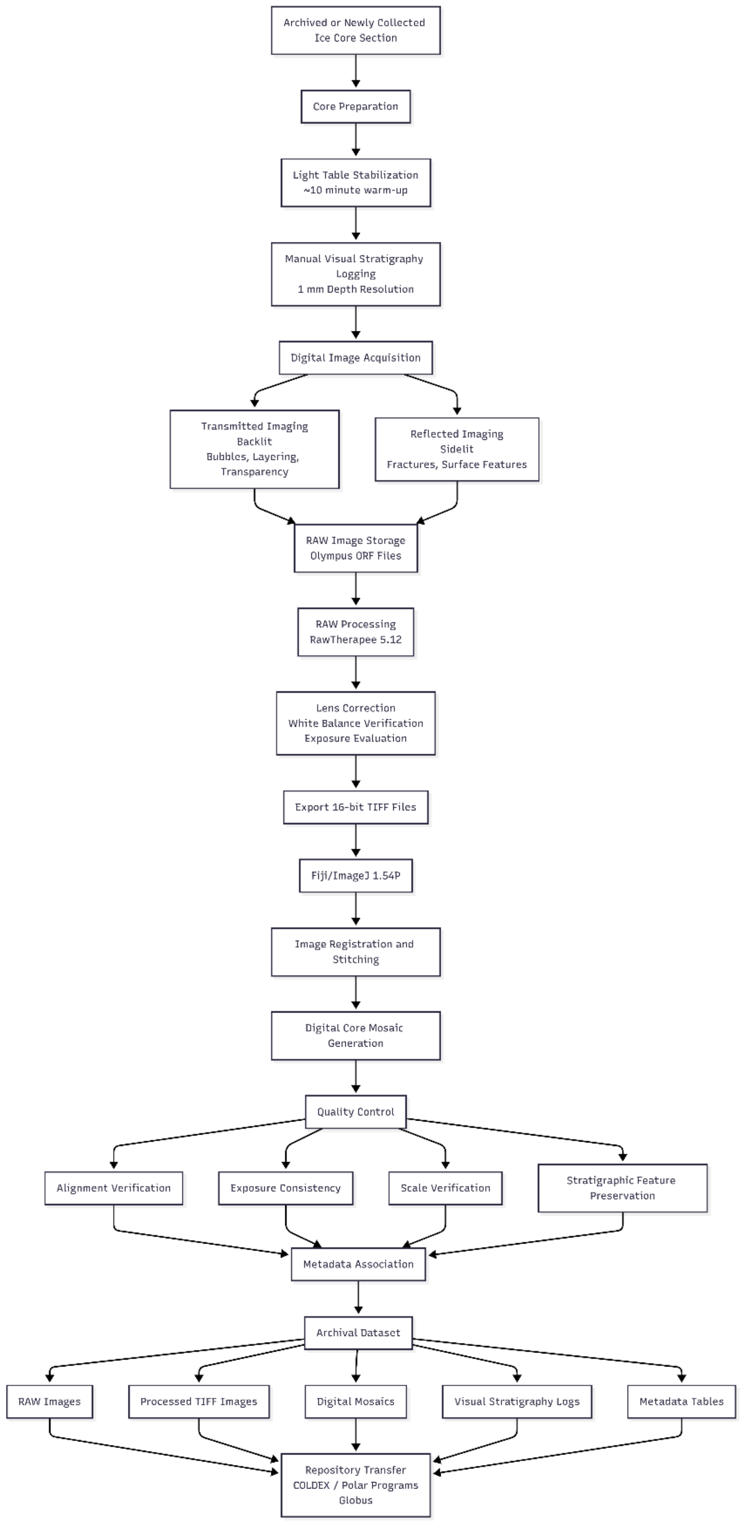

Figure 1. Overview of Visual Stratigraphy Imaging Workflow (image created in Mermaid.ai)

Ice Core Preparation and Visual Stratigraphy Documentation

Core Preparation

Archived ice core sections were removed from storage and prepared for visual examination and imaging.

Core sections were positioned with the flat surface facing upward on the imaging stage. Core orientation was maintained throughout documentation and imaging to preserve spatial relationships between physical observations and digital products.

Prior to imaging, the fluorescent light table was allowed to warm for approximately 10 minutes to stabilize illumination conditions.

Manual Visual Stratigraphy Logging

Visual stratigraphy was documented using standardized one-meter core logs with observations recorded at a 1 mm depth resolution.

Manual logging was completed before image processing to provide an independent record of visible ice properties.

Recorded information included:

- Core identification
- Core section number
- Tube number
- Depth interval
- Feature type
- Feature description
- Approximate size
- Shape and orientation
- Distribution
- Additional notes

Table 1. Visual Stratigraphy Logging Fields

Field	Description
Core ID	Ice core identifier
Section number	Physical core section
Tube number	Storage tube
Depth	Depth interval
Feature type	Bubble, fracture, inclusion, dust, etc.
Size	Approximate dimensions
Shape	Morphology
Orientation	Dip/elongation
Distribution	Spatial occurrence
Notes	Additional observations

Documented Ice Features

The following visual characteristics were documented.

Inclusions

Recorded characteristics included:

- Size
- Shape
- Distribution
- Relative abundance
- Relationship with surrounding ice structures

Bubble Structure

Bubble observations included:

- Bubble size
- Bubble shape
- Bubble abundance
- Distribution
- Elongation
- Dip direction (if any)

Fractures

Fractures were documented by:

- Depth location
- Orientation
- Length
- Extent
- Relationship with surrounding features

Core Handling Features

Features related to recovery and preparation were documented:

- Coring tool marks
- Saw marks
- Spalls
- Surface breaks

Optical Properties

Observed optical characteristics included:

- Cloudiness
- Transparency variation
- Clear/cloudy ice transitions
- Changes in optical appearance

Particulate Features

Visible particulate layers were documented including:

- Dust horizons
- Potential tephra layers
- Layer thickness
- Color
- Distribution

Imaging System and Equipment

Camera System

Digital images were collected using an Olympus C-8080 Wide Zoom digital camera with the factory-installed Olympus lens.

Images were collected in Olympus RAW (.ORF) format to preserve original sensor information.

The camera was operated in manual mode with fixed settings.

Table 2. Camera Acquisition Parameters

Parameter	Setting
Camera	Olympus C-8080 Wide Zoom
Lens	Factory Olympus lens
Format	ORF RAW
Resolution	3264 × 2448 pixels
Bit depth	12-bit
Aperture	f/5.6
ISO	50
Focal length	7 mm
Exposure compensation	0 EV
Focus	Manual
Delay	2 seconds
Height	14 inches

Image Acquisition

Illumination Configuration

Two imaging approaches were used.

Transmitted Imaging (Backlit)

Purpose:

- Bubble structure
- Internal layering
- Transparency variation
- Internal inclusions

Settings:

- Fluorescent light table
- Exposure: 1/25 second
- ISO: 50

Reflected Imaging (Sidelit)

Purpose:

- Surface features
- Fractures
- Relief features

Settings:

- Side illumination
- Exposure: 1/8 second
- ISO: 50

Image Capture Procedure

Each core section was photographed using four overlapping images.

Image positions:

Table 3. Image Acquisition Positions (camera alignment position).

Image	Position
1	15 cm
2	40 cm
3	65 cm
4	90 cm

During acquisition:

- Camera height remained fixed.
- Focus remained fixed.
- Exposure settings remained fixed.
- Two-second timer minimized vibration.
- Backlit and sidelit images were collected separately.

Image Calibration References

Calibration references included:

- Stouffer T2115 transparent 21-step grayscale wedge
- White reference
- Black reference
- Wooden metric ruler

The grayscale wedge was used for:

- White balance verification
- Exposure evaluation
- Future intensity normalization

The ruler provided:

- Spatial reference
- Mosaic alignment verification

RAW Image Processing

Processing was completed using RawTherapee 5.12.

Figure 2. Image Processing Workflow (image created in Mermaid.ai).

Lens Correction

The Olympus C-8080 camera profile available within RawTherapee was applied before image stitching.

Correcting lens distortion before stitching reduces geometric inconsistencies between overlapping images.

Exposure

Images were initially processed at 0 EV.

Histogram evaluation was used to identify images requiring adjustment.

Future improvements will incorporate grayscale-based exposure normalization.

Image Enhancement

To preserve natural ice appearance:

Disabled:

- Sharpening
- Noise reduction
- Color enhancement

Export

Processed images were exported as 16-bit TIFF files for Fiji/ImageJ processing.

Fiji/ImageJ Mosaic Generation

Processing Environment

Image stitching was completed using:

Fiji/ImageJ Version 1.54P

Stitching Workflow

Figure 3. Image Stiching Workflow (image created in Mermaid.ai).

Table 4. Fiji Processing Parameters (parameter unless noted).

Parameter	Setting
Fiji version	1.54t
Plugin	Grid/Collection Stitching
Grid size	Y=1, X=4*
Fusion method	row-by-row
Overlap	30 %*
Subpixel accuracy	On
Output format	Fuse and Display

Quality Control and Validation

Quality control evaluated:

- Exposure consistency
- Image alignment
- Spatial scale
- Preservation of stratigraphic features

Checks included:

- Ruler continuity
- Feature alignment
- Absence of duplicated structures
- Absence of major distortion
- Consistent illumination

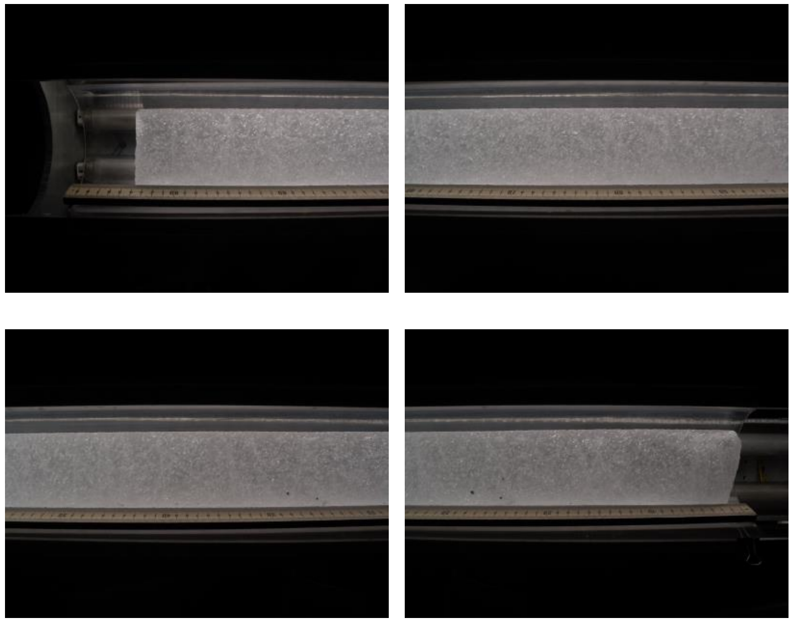

Figure 4. Individual Raw Files of Section 19 AH2416.

Figure 5. Stitched Mosaic of Section 19 after Quality Control and Validation.

Data Management and Archiving

File Structure

- AH2416
  - Field Log
  - Section 01*
    - Visual Stratigraphy
    - Digital Photography
      - Raw
        - Backlit
        - Sidelit
      - Tiff
        - Backlit
        - Sidelit
      - Mosaics
      - Calibration
      - Metadata

Figure 6. Recommended Data Organization

Metadata

Metadata spreadsheet should include:

Table 5. Metadata Structure

Field	Description
Core ID	
Section number	
Tube number	
Depth	
Image number	
Illumination	
Processing stage	
Notes	

Backup and Transfer

Data should be maintained in multiple locations:

- Processing computer
- External storage
- Cloud storage

Final datasets prepared for NSF COLDEX/Polar Programs transfer should include:

- RAW images
- TIFF images
- Digital mosaics
- Core logs
- Metadata
- Workflow documentation

Data transfer may be completed using Globus.

Future Improvements

Future workflow development includes:

- Automated grayscale normalization
- Improved image registration
- Automated feature detection
- Integration with HSI and ECM datasets
- Expanded application to newly collected cores

Conclusion

This workflow provides a standardized, low-cost approach for creating digital visual stratigraphy records of ice cores. By combining manual observations, standardized imaging, open-source image processing, and structured metadata management, the workflow produces reproducible digital records that support long-term ice core preservation and future scientific analysis.

Workflow Development Status

Version 1.0 — Initial Release

This workflow was developed during the NSF COLDEX Research Experience for Undergraduates (REU) program using archived ice core sections from AH2416, Allan Hills Blue Ice Area, Antarctica.

Version 1.0 establishes a reproducible framework for digital visual stratigraphy documentation, including standardized image acquisition, RAW image processing, digital mosaic generation, visual logging, quality control, metadata organization, and archival preparation.

Future updates will focus on improving image calibration procedures, including grayscale-based radiometric normalization, refinement of image registration and stitching methods, increased processing automation, and integration with complementary ice core datasets such as hyperspectral imaging (HSI), electrical conductivity measurements (ECM), and geochemical analyses.

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